

Human Motion Recognition Using IMU Sensors

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Abstract

This project investigates the use of Inertial Measurement Unit (IMU) sensors for human motion recognition through deep learning-based classification. Multichannel acceleration data collected from wearable IMU sensors is processed using a Convolutional Neural Network (CNN) architecture to classify various human activities. The proposed system is implemented using MATLAB and Simulink and is designed to achieve high classification accuracy while maintaining scalability for real-time, gesture-controlled applications. Experimental results demonstrate that the proposed approach effectively recognizes both static and dynamic activities.

I. INTRODUCTION

Human motion recognition plays a critical role in modern applications such as human-computer interaction, rehabilitation systems, wearable healthcare monitoring, and sports analytics. Accurate recognition of human activities enables intelligent systems to adapt to user behavior and provide context-aware responses.

This project aims to develop a classification model capable of recognizing patterns of human movement using data acquired from IMU sensors. These sensors record inertial signals that reflect body dynamics during different activities, resulting in multichannel time-series data. One of the key challenges in IMU-based activity recognition lies in handling noisy signals, sensor placement variability, and class imbalance while maintaining generalization to unseen data.

The primary objective of this work is to design a robust Human Activity Recognition (HAR) system that learns discriminative temporal features directly from raw inertial signals, eliminating the need for manual feature engineering. By leveraging deep learning techniques, the proposed approach seeks to achieve high accuracy while remaining suitable for real-time deployment in wearable and gesture-controlled systems.

II. LITERATURE

Using inertial sensors for Human Activity Recognition (HAR) is a growing area of study. Current uses range from medical rehabilitation and fall detection to VR/AR interaction and robotic control. IMU-based systems are helpful because they use little power, are small, have high sampling rates, and do not depend on outside equipment like cameras. Early HAR systems mostly used hand-designed feature extraction. Researchers measured features in the time domain (mean, variance, zero-crossings) or frequency domain (FFT coefficients, spectral entropy). Bao & Intille (2004) in [1] showed that even basic features could be used to classify daily activities like walking, sitting, standing, and climbing stairs using machine learning like SVM and decision trees. As bigger IMU datasets (like PAMAP2, WISDM, Opportunity) became available, deep learning became popular. Yang et al. (2015) in [2] introduced one of the first HAR models using CNNs, processing IMU data directly without needing pre-designed features. Later work expanded on this using stacked CNNs, LSTMs, and CNN-LSTM networks for modeling over time. Hammerla et al. (2016) in [3] compared deep architectures and found that CNNs did better than traditional ML models across different wearable datasets.

Recent studies highlight three main areas:

- 1) **Representation Learning:** Autoencoders, contrastive learning, and transfer learning are used to learn features that are consistent regardless of motion or person.
- 2) **Windowing and Segmentation Strategies:** Window length and overlap affect how well the system performs. Usually, 1 to 2-second windows with 25 to 50% overlap work best for repetitive motions.
- 3) **Multimodal Fusion:** Combining data from accelerometers, gyroscopes, and magnetometers can improve recognition accuracy, especially for complex actions.

The studies by Demrozi et al. (2020) in [4] and Gomaa & Khamis (2023) in [5] discuss the challenges of different sensor placements and generalization across different people. Rehman et al. (2024) in [6] note the need for good validation and transfer learning to handle noise, drift, and inconsistent sampling.

This project is builded on this work by implementing a CNN-based classification process for fixed-length windows, z-score normalization, and category filtering. Our system is designed for a controlled IMU dataset and aims to be used for real-time applications using MATLAB's imuSensor and hardware integration.

III. DATA COLLECTION & PREPROCESSING

A. Dataset Structure

The dataset consists of CSV files that each have 8 entries:

- Timestamp of the recorded measurement
- 3 entries for measurements from 3-axis accelerometer placed on the lower-back (back_x, back_y, back_z)
- 3 entries for measurements from 3-axis accelerometer placed on the thigh (thigh_x, thigh_y, thigh_z)
- Activity labels for the motions

B. Preprocessing Pipeline Before Model Training

Before training the model, the raw IMU data is processed as follows:

- 1) **Time Synchronization & Format Conversion:** All CSV files are processed using MATLAB's *datastores*, which supports out-of-memory processing for large datasets, removing the need to have all the data loaded into memory. Data types are converted to double precision for all numerical fields.
- 2) **Segmentation with Sliding Windows:** A widely-used segmentation strategy in HAR:
 - Window length: 500 samples
 - Overlap: 250 samples (50)

Each window produces one training sample (sequence). This method captures both temporal continuity and local motion characteristics, consistent with previous research recommending 1 to 2 second windows.

- 3) **Label Filtering (Ignored Classes):** One drawback of the dataset is that some labels have insufficient data to train a model. Therefore, the labels with insufficient data will not be used for training or testing. These labels are classified as 'ignored labels'.
- 4) **Normalization:** Each window undergoes z-score normalization computed individually per sequence:

$$X_{norm} = \frac{X - \mu}{\sigma}$$

The effects of this are:

- Differences in sensor placement have less impact
- Amplitude variations do not dominate
- Gradient-based learning is stabilized

IV. SOLUTION STRATEGY

A. Deep Learning Architecture

The model implements a multi-layer 1D Convolutional Neural Network (CNN) tailored for sequential IMU data. Rather than treating data points in isolation, the architecture is designed to capture the temporal "flow" of movement.

- **Feature Extraction:** `Conv1D` layers identify motion patterns, while batch normalization ensures training stability across different sensor scales.
- **Hierarchical Complexity:** By increasing filter depth from $32 \rightarrow 64 \rightarrow 128$, the network progressively captures more complex motion transitions.
- **Noise Reduction:** `MaxPooling` layers downsample the features to reduce dimensionality and filter out signal noise.
- **Generalization:** `Dropout` and `Global Average Pooling` are used to prevent overfitting, ensuring the model generalizes well to new subjects while keeping the parameter count efficient.
- **Classification:** A final `Softmax` layer provides the probability distribution for each activity class.

B. Training Configuration

The training process is optimized to handle the inherent noise and class imbalances typical of sensor-based datasets:

- **Optimizer:** Adam, utilizing a learning rate decay every 25 epochs to refine weights as the model converges.
- **Duration:** 75 epochs with a batch size of 16.
- **Objective Metric:** The *F-score* is used as the primary metric to ensure performance is maintained even across imbalanced activity classes.
- **Hardware:** GPU acceleration is enabled to handle the computational demands of the 1D convolutions.

C. Evaluation & Testing

The evaluation workflow follows modern HAR (Human Activity Recognition) standards to support rigorous error analysis:

- **Prediction Pipeline:** Testing involves reconstructing the IMU windows and processing them via `minibatchpredict` to generate score vectors.
- **Label Mapping:** Score vectors are mapped to categorical labels to calculate overall accuracy.
- **Error Analysis:** A Confusion Matrix is generated to visualize classification performance, allowing for a detailed look at which specific activities (e.g. "sitting" vs. "standing") the model may be confusing.

V. RESULTS

A. Training Performance

Figure 1 illustrates the training loss progression over iterations. The loss decreases rapidly during the initial training phase, indicating effective learning of low-level motion features. After approximately 6000 iterations, the loss stabilizes around 0.1, suggesting convergence of the model. Minor oscillations observed throughout training are expected due to minibatch-based optimization and do not indicate instability. Overall, the loss trend demonstrates stable and successful training behavior without signs of divergence.

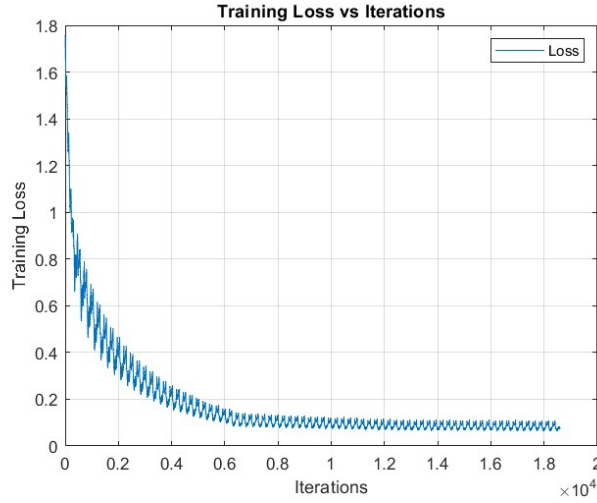


Fig. 1. Training Loss

Figure 2 presents the training accuracy as a function of training iterations. The accuracy increases rapidly during the early stages of training, reflecting the model's ability to quickly learn discriminative motion features. After approximately 6000 iterations, the accuracy exceeds 95% and gradually stabilizes around 98–99%, indicating convergence. Minor fluctuations in accuracy are expected due to minibatch-based optimization and do not reflect instability. The overall trend confirms effective and stable learning behavior.

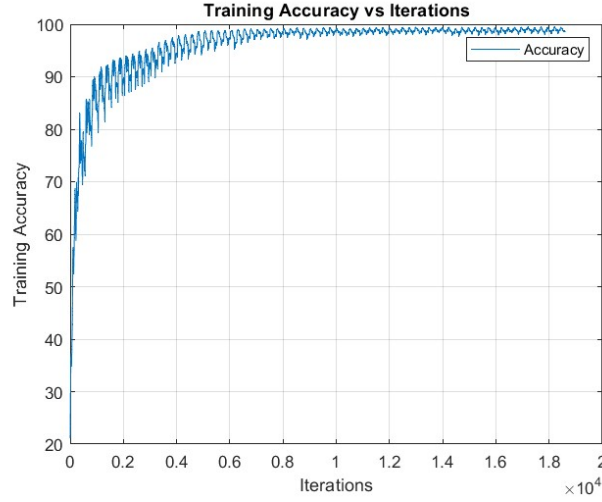


Fig. 2. Training Accuracy

B. Test Performance

The performance of the proposed CNN model is illustrated through the confusion matrix in Figure 3 and the detailed statistical breakdown in Table I. The system achieves an average accuracy of 96.74% and an average F1-score of 0.92. As shown in Figure 3, the model demonstrates near-perfect discriminative capability for "Lying," which was classified with 100% accuracy. Table I corroborates this, reporting a perfect 1.00 F1-score for the "Lying" category. This suggests that the spatial orientations captured by the sensors provide a distinct feature set for horizontal postures. Similarly, "Sitting" displays high reliability with a precision of 0.99, indicating that the model rarely produces false positives for this state. However, Figure 3 reveals a notable misclassification between "Walking" and "Standing." Specifically, 69 instances of walking were misclassified as standing, which is reflected in Table I by the lower recall of 0.82 for the walking activity. This confusion is likely caused by the brief pauses while walking, which mimic the signal profile of a standing posture. This misclassification also reduces the precision of "Standing" to 0.78. Despite these localized challenges in differentiating between similar upright postures, the overall high metrics across both Figure 1 and Table I indicate a robust architecture capable of effective human activity recognition.

	standing	walking	sitting	lying	running
standing	302				
walking	69	363	4		4
sitting			294		18
lying				197	
running	16				94
	standing	walking	sitting	lying	running

Fig. 3. Confusion Matrix of Test Dataset

TABLE I
SYSTEM PERFORMANCE METRICS

Activity	Accuracy (%)	Precision	Recall	F1-Score
Standing	93.75	0.78	1.00	0.88
Walking	94.74	1.00	0.82	0.90
Sitting	98.38	0.99	0.94	0.96
Lying	100.00	1.00	1.00	1.00
Running	97.21	0.81	0.85	0.83
Average	96.74	0.92	0.92	0.92

VI. CONCLUSION

In this project, a deep learning-based classification model using 1D Convolutional Neural Networks (CNN) was successfully developed to recognize human motion patterns from IMU sensor data collected from the lower back and thigh. The implementation of a robust preprocessing pipeline, including time synchronization, 500-sample sliding window segmentation, and z-score normalization, effectively minimized the impact of sensor placement variations and signal noise.

The experimental results demonstrate the high efficacy of the proposed architecture: The system achieved a 96.74% overall average accuracy and an F1-Score of 0.92.

Class-specific performance analysis showed near-perfect recognition for distinct static postures, with 100% accuracy for 'Lying' and 98.38% for 'Sitting'.

Confusion matrix analysis revealed that the primary challenge lay in distinguishing between dynamically similar activities like 'Standing' and 'Walking'; however, the model still maintained an accuracy above 93.75% for these classes.

This project confirms that processing raw inertial data through a specialized CNN architecture eliminates the need for manual feature extraction while delivering high-precision results. Future work will focus on deploying this model for real-time applications using MATLAB's `imuSensor` framework and expanding the dataset to improve generalization across different users and hardware configurations.

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